CS-Net: Deep Multibranch Network Considering Scene Features for Sharpness Assessment of Remote Sensing Images

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***Abstract*— Recently, many new results of sharpness assessment for digital images have been achieved, which help to select valuable images from massive images with ragged quality. However, remote sensing images encompass a wide range of scenes with diverse characteristics, and their acquisition is often influenced by blurs and noises. Many commonly used sharpness assessment methods based on uniform metrics face challenges in ensuring both subjective and objective impartiality when applied to remote sensing images. Therefore, a novel method for assessing the sharpness of remote sensing images based on a deep multibranch network considering scene features is proposed. In the method, a multitask module (MTM), comprising scene classification and sharpness assessment tasks, is proposed to comprehensively consider the potential impact of diverse scene characteristics on the assessment of sharpness. The accuracy of sharpness assessment is improved by sharing features that reflect the correlation between the scene and sharpness. To overcome the issue of imbalanced task predictions during the joint training of multiple tasks, a total loss function using gradient balance strategy is designed. In addition, the improved attention module (IAM) and the feature fusion module (FFM) are used to better utilize feature information at different scales. Experimental results obtained from datasets demonstrated that the proposed method can outperform the existing comparable methods and achieve satisfactory results, proving its feasibility and effectiveness.**

***Index Terms*— Feature sharing, gradient balance strategy, remote sensing images, scene classification, sharpness assessment.**

I. Introduction

THE data volume of remote sensing imagery has been experiencing rapid growth in recent times, promoting its widespread utilization across various applications, such as environmental monitoring [1], disaster analysis [2], and resource investigation [3]. Remote sensing imagery constitutes the cornerstone of these applications. However, the quality of remote sensing images is often impacted by imaging systems [4] and meteorological factors [5], resulting in the loss of valuable information within the images. Simultaneously, these unavailable remote sensing images consume substantial storage space and computing resources [6]. Therefore, it is necessary to assess the quality of remote sensing data to provide available data for numerous applications from voluminous remote sensing images.

There are numerous metrics for evaluating image quality, among which sharpness holds significant importance [7]. In the field of remote sensing, sharpness refers to the clarity of object edges and the level of detail within an image. It directly influences the interpretability and information retrieval capability of remote sensing imagery. Furthermore, as the resolution of remote sensing imagery continues to improve, more detailed information regarding features and edges of objects can be captured. However, images may suffer from higher levels of noise, blurring, and other undesirable artifacts due to factors such as sensor characteristics, focusing accuracy, and motion blur. The blurring not only occurs in satellite remote sensing images [8] but also in aerial remote sensing images [9], significantly affecting the quality of these images. Hence, an accurate assessment of image sharpness can provide a precise foundation for determining image quality.

Numerous studies on image sharpness assessment have been conducted. According to whether human involvement is present, methods can be categorized into subjective and objective approaches. Subjective assessment methods primarily rely on human visual perception to evaluate image sharpness. Objective assessment methods include spatial-domain-based, frequency-domain-based, and machine learning-based methods.

Spatial-domain-based methods evaluate image sharpness by extracting spatial features such as gradient and edge information [10], [11]. The gradient is extracted by means of a greyscale gradient function, which assesses the sharpness of an image by measuring the difference between neighboring pixels [12]. Classical gradient functions include the Tenengrad function [13], Laplacian function [14], Brenner function [15], sum of modulus of gray difference (SMD) [16], and energy function [17]. In addition to these well-established gradient-based evaluation methods, several novel approaches have been investigated. Zhang et al. [18] assessed edge sharpness by computing the grayscale values in eight directions of each pixel. Wang et al. [19] used the block standard deviation sharpness assessment method based on

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spatial domain to evaluate the image quality of 208 GF-4 panchromatic images in Zhengzhou and Chifeng. The edge detection function is also commonly employed in the evaluation of image sharpness [20], with popular algorithms including the Canny operator [21], the Sobel operator [22], and others. For instance, Liu et al. [23] utilized the Canny edge detection algorithm to obtain the positions and orientations of image edges and derived a sharpness assessment metric by averaging the weighted edge widths. This method has demonstrated a high level of accuracy. Pampanoni et al. [24] proposed a flexible and fully automated sharpness evaluation method based on edge analysis. This method is applicable to panchromatic, multispectral, and hyperspectral images, as well as images with various resolution ranges.

Frequency-domain-based methods evaluate image sharpness by transforming the high- and low-frequency components in the frequency domain, which represent the sharp and blurred regions of the image [25], [26]. Commonly used evaluation functions include Fourier transform [27], [28] and wavelet transform [29], [30]. In addition, De and Masilamani [31] presented an image sharpness measure that uses multivariate adaptive regression splines for predicting the quality of an image corrupted by image blur using discrete cosine transform statistics. Baig et al. [32] proposed a no-reference image quality assessment method based on discrete Fourier transform (DFT). This approach calculates the image derivative and block-based DFT and then combines them to obtain a sharpness measure. The wavelet transform function enables the utilization of localized features within an image to obtain the assessment results of the imagery. Gvozden et al. [26] presented a fast blind image sharpness/blurriness assessment model, which generates local contrast image maps by computing the root-mean-squared values for each image pixel within a defined size of local neighborhood. Wang et al. [33] proposed a no-reference stereoscopic image quality assessment model using heterogeneous ensemble learning "quality-aware" features from luminance image, chrominance image, disparity, and cyclopean images via quaternion wavelet transform. Xiang et al. [34] proposed a reduced-reference light field image quality assessment model based on pseudo light field image and 4-D wavelet transform, which achieves good results.

Machine learning-based methods for image sharpness assessment [35] have primarily relied on support vector regression and support vector clustering techniques. Pei et al. [36] proposed a sharpness evaluation method based on large-scale structures. This method calculates the statistics proportion of edges with different widths and the mean value of the maximum gradients from the 51st to the 250th as the features of the image, which can avoid the interference of small textures and contrast around the edge to the evaluation results. Liu et al. [37] developed an effective blind image blur assessment model based on a novel orientation-aware local pattern operator, which fully considers the impact of anisotropy of orientation selectivity mechanism and the gradient orientation effect on visual perception. With the continuous development of deep learning methods, an increasing number of approaches based on deep learning have been employed for image sharpness assessment [38]. For instance, Li et al. [39] employed a deep convolutional neural networks (CNNs) to extract high-level features from image patches and obtained sharpness assessment values using the least-squares regression model. Liu et al. [40] proposed a no-reference image quality assessment approach that learns from rankings for addressing the problem of limited image quality assessment dataset size. Zhu et al. [41] proposed a no-reference image quality assessment metric based on deep meta-learning, the underlying idea of which is to learn the meta-knowledge shared by human when evaluating the quality of images with various distortions This learned meta-knowledge can then be easily adapted to unknown distortions. Yang et al. [42] presented the multidimension attention network for no-reference image quality assessment (MANIQA) to improve the performance on generative adversarial networks (GANs)-based distortion.

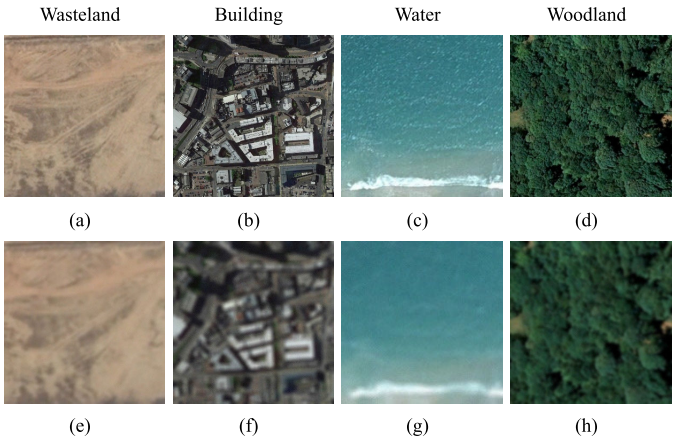


Fig. 1. Remote sensing imagery of various scenes with different levels of sharpness. (a)–(d) Clear images of the four scenes (wasteland, building, water, and woodland). (e)–(h) Degraded images of the four scenes (wasteland, building, water, and woodland).

However, numerous challenges still persist in evaluating the sharpness of remote sensing imagery. Remote sensing images cover a wide range of areas, featuring intricate and diverse scenes that are rich in texture information. Furthermore, the acquisition of remote sensing images is often affected by blurs and noises [43]. Many existing methods fail to adequately consider the effects of scene characteristics, blurs, and noises. Moreover, it is difficult to ensure the unbiasedness of these existing assessment methods based on unified indexes when dealing with remote sensing images. For instance, Fig. 1 shows randomly selected remote sensing images of four scenes: wasteland, building, water, and woodland. Among them, Fig. 1(a)–(d) represents clear images of the four scenes, while Fig. 1(e) and (h) represents the corresponding degraded images. Five commonly used sharpness assessment metrics (Tenengrad function, Laplacian function, Brenner function, SMD, and energy function) were employed to assess the sharpness of these images.

The Tenengrad function assesses the sharpness of an image by computing the gradient for each pixel and accumulating the squared gradient values. A higher value of the Tenengrad function indicates a clearer image. The Laplacian function is commonly used to compute the local variations in an image. A clear image typically possesses more edges and details, hence leading to more pronounced variations among its local

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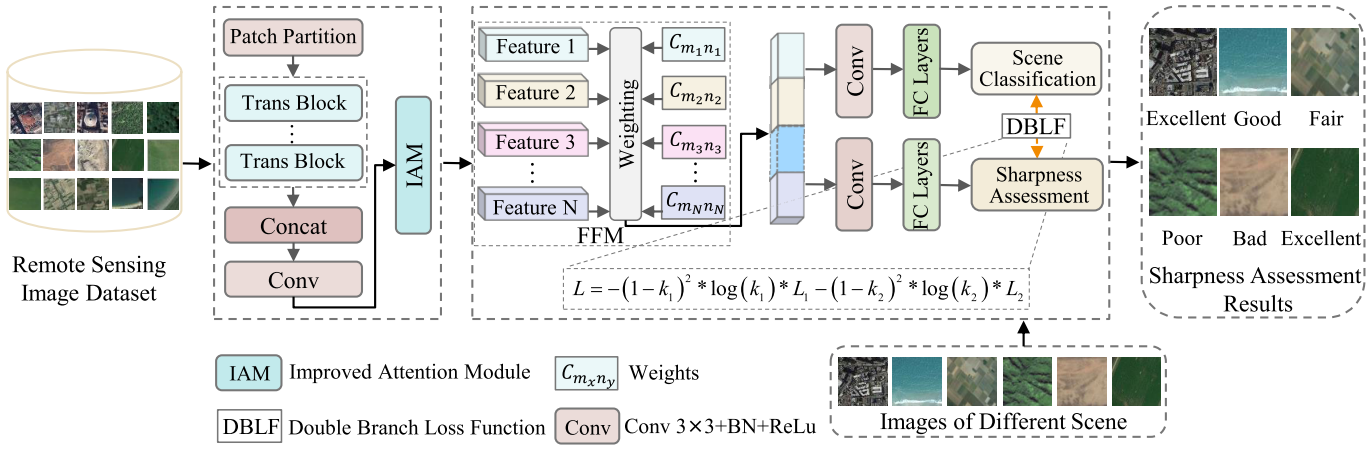


Fig. 2. Overall architecture of CS-Net.

pixel values. The Brenner function calculates the difference in pixel values within a specific area of an image to reflect the edge information present in the image. A higher Brenner function value indicates richer edge information within the image region, meaning that the image is clearer. SMD calculates image sharpness by summing grayscale differences between adjacent pixels. Higher SMD means clearer images with significant variation, while lower SMD suggests potential blurriness or less detail due to minimal grayscale differences. The energy function assesses pixel differences or changes in an image by weighting attributes such as gradients and contrasts. It indicates image sharpness, with higher values denoting clearer images.

The results of the five traditional sharpness assessment metrics in Fig. 1(a)–(d) show significant differences. For instance, the Tenengrad value in Fig. 1(a) is 4.88, whereas it reaches 1491.71 in Fig. 1(b). This might be due to significant variations in the texture features of remote sensing images from different scenes that are subjectively perceived as having similar sharpness, affecting their sharpness assessment. Analyzing Fig. 1(e)–(h), it becomes evident that as sharpness decreases, the differences between the sharpness assessment scores among different scenes diminish but persist. For instance, the Laplacian value in Fig. 1(e) is 75.69, while in Fig. 1(g), it stands at 454.73. Moreover, some metrics demonstrate inaccuracies in assessing the image sharpness of scenes with weak textures. An illustrative case is the comparison between Fig. 1(c) and (g), where the sharpness of (c) surpasses that of (g). Nonetheless, both its SMD value (19.19) and energy value (58.26) are lower than the corresponding SMD value (32.45) and energy value (90.04) of Fig. 1(g). Consequently, unified indices might not precisely depict the varying levels of sharpness in remote sensing images across diverse scenes.

To address the aforementioned issues, a novel method based on the deep multibranch network considering the scene features (CS-Net) is proposed for the sharpness assessment of remote sensing images. The main contributions are summarized as follows.

1) A multitask module (MTM) is proposed to comprehensively consider the impact of scene diversity and disparities in texture features on the assessment of sharpness in remote sensing images. MTM divides the evaluation of remote sensing images into two tasks: scene classification and sharpness assessment. Features associated with both scene classification and sharpness assessment can be shared through multitask learning.
2) A double-branch loss function (DBLF) is designed to determine the optimal weights for the scene classification loss and sharpness assessment loss. DBLF alleviates potential imbalances in losses that may arise during the simultaneous training of multiple tasks and reduces the significant training overhead caused by manual weight exploration. In addition, an improved attention module (IAM) is utilized to sharpen the network's focus on crucial information, and a feature fusion module (FFM) is employed to leverage the feature information at different scales.
3) Extensive experiments on the sharpness assessment dataset verified the feasibility and effectiveness of the presented method. The dataset was created based on AID dataset (a publicly available remote sensing dataset), which consists of six scene category labels and five sharpness level labels.

II. Methodology

The CS-Net proposed in this article can better utilize scene information through multibranch learning and enhance the utilization of crucial feature information by employing an improved IAM, resulting in improved accuracy for the sharpness assessment of remote sensing images. The traditional methods of sharpness assessment lack stability in evaluating remote sensing images across different scenes. In fact, there are significant differences in texture features among remote sensing images in different scenarios. Therefore, the proposed method explores the inherent relationship between scene texture and image sharpness, thereby assessing the sharpness of remote sensing images while considering the scene category.

A. Overall Architecture of CS-Net

The overall architecture of CS-Net is illustrated in Fig. 2. Initially, a multiscene dataset of remote sensing, comprising

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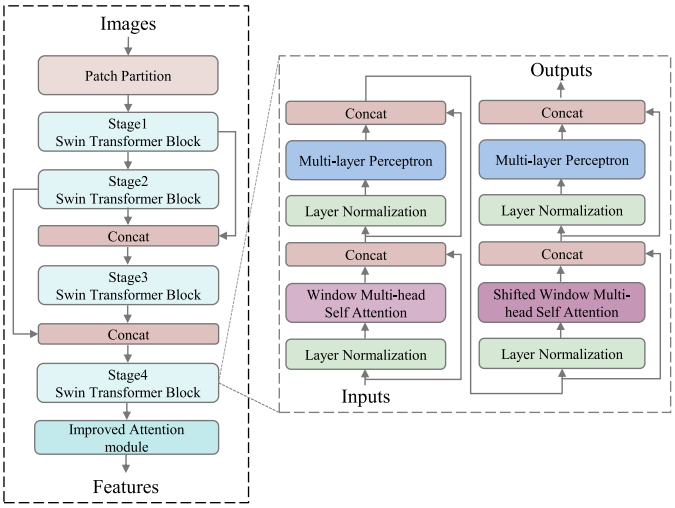


Fig. 3. Illustration of feature extraction.

six scene types, is established in accordance with China's remote sensing monitoring land use cover classification system. To preliminarily extract convolutional features from remote sensing images, a modified Swin-transformer [44] is employed as the backbone network. These features undergo processing by the IAM to derive feature maps (FMs). Subsequently, the amalgamated FMs are simultaneously fed into the scene classification module (SCM) and the sharpness assessment module (SAM) for model training. During the multitask training process, DBLF is utilized to determine the optimal weights for integrating the SCM loss and the SAM loss within the joint loss function. Finally, sharpness assessment results of remote sensing images can be attained.

B. Feature Extraction Using IAM

Transformers exhibit inherent advantages in global feature modeling [45]. Key features between image sharpness and the scene can be more accurately captured through global features. Effective feature extraction largely determines the results of remote sensing image scene classification and sharpness assessment. Therefore, the Swin-transformer was adopted for feature extraction. Meanwhile, an IAM is proposed to enable networks to better understand the importance of different channels and positions within images, thereby focusing more on capturing essential feature information and reducing the influence of irrelevant background information.

Transformer is initially designed for machine translation tasks in the field of natural language processing, and it has achieved remarkable performance [46]. Subsequently, the vision transformer (ViT) model was proposed, which directly applies the transformer to the domain of image classification [47]. To address the high computational complexity of ViT, the Swin-transformer model was presented.

The feature extraction module proposed in this paper is shown in Fig. 3, which consists of a patch partition module, four stages, and an improved feature attention module. The first patch partition module reduces the dimensionality of the input images. The number of Swin-transformer blocks

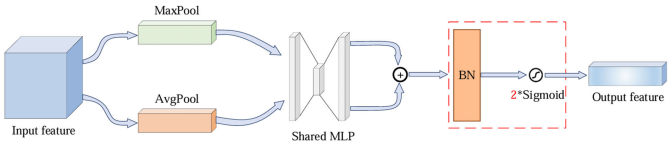


Fig. 4. Processing flow of the improved channel attention module.

in the four stages is 2, 2, 6, and 2. The Swin-transformer block consists of layer normalization, window multihead self-attention, shifted window multihead self-attention, and multilayer perceptron (MLP). In addition, skip connections are utilized to ensure the preservation of complete image information. The multilevel features from stages 1, 2, 3, and 4 extracted by Swin-transformer are $56 \times 56 \times 256$, $28 \times 28 \times 512$, $14 \times 14 \times 1024$, and $7 \times 7 \times 2048$ in size, respectively. To maximize the use of these four different scales of features, after processing by the IAM, FMs FM 1, FM 2, FM 3, and FM 4 with sizes of $56 \times 56 \times 128$, $28 \times 28 \times 256$, $14 \times 14 \times 512$, and $7 \times 7 \times 1024$ are obtained.

To make the feature extraction network more focus on important information and reduce the influence of useless background information, the convolutional block attention module (CBAM) attention mechanism [48] is introduced. Moreover, an improvement is made to the CBAM, aiming to eliminate any potential impact on the initial parameters of the pretrained model. The CBAM attention mechanism consists of two modules: the channel attention module and the spatial attention module. For a given feature, the CBAM attention mechanism infers the attention weight matrix along both channel and spatial dimensions. Then, the feature is adaptively adjusted by multiplying the attention weight matrix with the original feature. By this means, the shallow and deep features of remote sensing images have been effectively utilized. In this article, in the original channel attention module of the CBAM attention mechanism, the batch normalization (BN) layer is added before the Sigmoid activation function and the sum of BN layer coefficients is initialized to 0. In addition, the Sigmoid function is multiplied by 2. The processing flow of the improved channel attention module is shown in Fig. 4.

Specifically, for a feature with dimension $H \times W \times C$, the channel attention module uses both maximum pooling and average pooling to compress the spatial dimension of the feature, thereby generating two distinct channel features. The maximum pooling solely considers the largest element while disregarding other elements in the region, which enables the retention of more texture features inherent in remote sensing images. On the other hand, the average pooling calculates the mean value of all elements within the region, thereby preserving more background features prevalent in remote sensing images. After the pooling operation, the dimensions of these two features are $1 \times 1 \times C$. Subsequently, they are input into the MLP two-layer neural network, which utilizes shared parameters. In order to reduce the number of parameters, the first layer's neuron count is set to C/R, and the second layer's neuron count is set to C, where R represents the compression ratio. The two features produced by the MLP are combined, and the attention weight matrix is acquired

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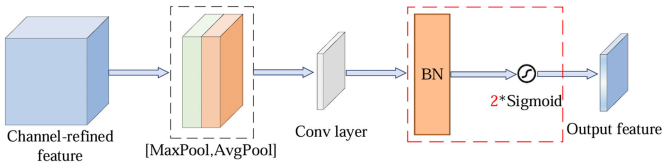


Fig. 5. Processing flow of the improved spatial attention module.

through the normalization layer and the Sigmoid activation function. Finally, the output feature from the channel attention module is obtained by performing element-wise multiplication between the attention weight matrix and the original feature. The calculation formula for the original channel attention weight matrix is as follows:

$$\begin{aligned} M_c(F_{\text{in}}) &= \sigma(\text{MLP}(\text{AvgPool}(F_{\text{in}})) + \text{MLP}(\text{MaxPool}(F_{\text{in}}))) \\ &= \sigma\left(w_1\left(w_0\left(F^c_{\text{avg}}\right)\right) + w_1\left(w_0\left(F^c_{\text{max}}\right)\right)\right) \end{aligned} \tag{1}$$

where F_{in} represents the original feature; F^c_{max} and F^c_{avg} represent the features output by maximum pooling and average pooling, respectively; w_1 and w_0 are the weight matrices of the MLP network; and σ represents the Sigmoid activation function.

After making the improvements, in the initial state of the network, the calculation formula of the original channel attention weight matrix is changed as follows:

$$\begin{aligned} M'_c(F_{\text{in}}) &= 2 * \sigma(\text{BN}(\text{MLP}(\text{AvgPool}(F_{\text{in}})) \\ &\quad + \text{MLP}(\text{MaxPool}(F_{\text{in}})))) \\ &= 2 * \sigma(0) \\ &= 1. \end{aligned} \tag{2}$$

The spatial attention module is a supplement to the channel attention module, which can highlight the feature information of the meaningful target position. The idea of improvement is similar to that of channel attention, and in the original spatial attention module of the CBAM attention mechanism, a BN layer with zero-initialized weights is incorporated before the Sigmoid activation function. Subsequently, the Sigmoid function is multiplied by the coefficient 2. The processing flow of the improved spatial attention module is shown in Fig. 5.

Specifically, two spatial features with dimensions $H \times W \times 1$ are generated by applying maximum pooling and average pooling to the feature with dimensions $H \times W \times C$. Subsequently, these two features are integrated using a merge operation. Finally, they are input into a convolution layer with a convolution kernel size of 7×7, and the spatial attention weight matrix is generated through the BN layer normalization and the Sigmoid activation function. The calculation formula for the original spatial attention weight matrix is as follows:

$$\begin{aligned} M_S(F_{\text{in}}) &= \sigma\left(f^{7*7}\left(\left[\text{AvgPool}(F_{\text{in}}), \text{MaxPool}(F_{\text{in}})\right]\right)\right) \\ &= \sigma\left(f^{7*7}\left(F^s_{\text{avg}}, F^s_{\text{max}}\right)\right) \end{aligned} \tag{3}$$

where F_{in} represents the feature output by the channel attention module; F^s_{max} and F^s_{avg} represent the features output by maximum pooling and average pooling, respectively; f^{7*7} represents the convolution operation with a convolution kernel of 7×7; and σ represents the Sigmoid activation function.

After making the improvements, in the initial state of the network, the calculation formula of the original spatial attention weight matrix is changed as follows:

$$\begin{aligned} M'_S(F_{\text{in}}) &= 2 * \sigma\left(\text{BN}\left(f^{7*7}\left(\left[\text{AvgPool}(F_{\text{in}}), \text{MaxPool}(F_{\text{in}})\right]\right)\right)\right) \\ &= 2 * \sigma(0) \\ &= 1. \end{aligned} \tag{4}$$

Then, in the initial state, the calculation formula of the improved attention mechanism is shown in the following equation:

$$\begin{aligned} \text{Output} &= M'_S\left(M'_c(F_{\text{in}}) * F_{\text{in}}\right) * M'_c(F_{\text{in}}) * F_{\text{in}} \\ &= F_{\text{in}}. \end{aligned} \tag{5}$$

From (5), in the initial state, the improved attention mechanism would output the original feature. Therefore, by this means, the initial parameters of the pretraining model are not changed, and the network gradient backpropagation in the initial state is not affected.

C. Sharpness Assessment Based on FFM

The MTM in CS-Net is mainly composed of two components: the SCM and the SAM. Due to the varying effects of FMs at different levels on the sharpness assessment results, giving equal importance to each FM may not accurately depict the contribution of individual feature levels to the outcomes. Therefore, the FMs from the output of the backbone network are fused using an introduced FFM. Subsequently, in the scene classification branch, a fully connected (FC) layer is devised to link the fused features, and the output nodes of the FC layer correspond to six distinct scene categories, namely, woodland, water, wasteland, cropland, building, and grassland. Finally, the output nodes output the probability value of each category, and the category with the highest probability value is selected as the final scene classification result. The sharpness assessment branch and the scene classification branch adopt similar ideas. The output nodes of the FC layer in the sharpness assessment branch correspond to five sharpness levels for the remote sensing images, namely, excellent, good, fair, poor, and bad. The sharpness level with the highest probability is then selected as the final sharpness assessment result for the image.

Different features play distinct roles in image sharpness assessment. Taking inspiration from the work of the researchers [49], an FFM is introduced to more effectively incorporate various features in the MTM. As shown in Fig. 6, within this FFM, it adaptively assigns different weights to each feature channel to enhance task performance. First, FMs from four different levels go through global average pooling (GAP) [50] and are then concatenated together as inputs. Afterward, the FC layers and the Sigmoid function are utilized to obtain the weights of each channel, which are directly multiplied and summed with the concatenated FMs. Next, the FC layers are designed to obtain scores for each scene category and each sharpness level.

D. DBLF-Based Loss Calculation

A DBLF based on a gradient balancing strategy is designed to evaluate the prediction loss of CS-Net. Meanwhile, the

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cross-entropy loss function is used to evaluate the prediction loss of the SCM and the SAM. The calculation formula of the cross-entropy loss function is as follows:

$$L_{1,2} = -\sum_{K=1}^{N}\sum_{I=1}^{C} y_{ki}\log\left(P_{y_{ki}}\right) \tag{6}$$

where L_1 represents the loss value of the SCM, L_2 represents the loss value of the SAM, y_{ki} is the label of whether sample k belongs to category i, $P_{y_{ki}}$ is the probability that the model predicts sample k belonging to category i, n is the total number of samples, and C is the total number of categories. The total loss function of CS-Net can be expressed as follows:

$$L = \lambda_1 L_1 + \lambda_2 L_2, \quad \lambda_{1,2} \in (0, 1) \tag{7}$$

where λ_1 represents the loss weight of the SCM and λ_2 represents the loss weight of the SAM.

In the joint training of scene classification and sharpness assessment tasks, the design of the total loss function is directly related to the training difficulty and the final performance of the network, so the definitions of weight factor λ_1 and λ_2 are very important. In the joint training of multitask learning, the gradient sizes of backpropagation may not be in the same order of magnitude when different tasks are trained. This discrepancy can lead to a situation where one task performs exceedingly well, while the other tasks almost fail. To address this issue, the gradient balancing strategy of focal loss [51] is employed. This strategy dynamically adjusts the weight values based on the accuracy performance of the two tasks during the training process. Its objective is to prioritize the more challenging tasks by decreasing the weight values of tasks with better accuracy performance and increasing the weight values of tasks with poorer accuracy performance.

The calculation expression of the loss weight is shown in the following equation:

$$\lambda_i = \mathrm{FL}(k_i, \gamma_i) = -(1 - k_i)^{\gamma_i} * \log(k_i), \quad i = 1, 2 \tag{8}$$

where k_i represents the evaluation index of the current training batch of each branch, which is the average correct rate of the current batch in this article, and $k_i \in [0, 1]$, and γ_i is generally taken as 2.

According to (8), the total loss function in (7) can be further expressed as follows:

$$L = -(1 - k_1)^2 * \log(k_1) * L_1 - (1 - k_2)^2 * \log(k_2) * L_2. \tag{9}$$

When k_i of the network is closer to 1, that is, the better the network performance, the smaller the weight value λ_i, these make the network training biased toward another task. When k_i of the network tends to 0, λ_i tends to infinity, which can make the network training completely biased to poorly performing tasks and bring the problem of gradient explosion. Therefore, in this article, focal loss is not used in the initial stage of network training, but only after ensuring that the network weights are updated to a certain degree before adopting this balancing strategy. Considering the importance of the two tasks and the difficulty of training, when the total number of training steps is less than 1000, this article intends to set the scene classification branch weight factor $\lambda_1 = 0.2$ and the quality evaluation branch weight factor $\lambda_2 = 0.8$. The gradient balancing strategy in (8) is used to dynamically adjust λ_1 and λ_2 when the total number of training steps is greater than or equal to 1000.

TABLE I
SCENES OF EXPERIMENT DATA AND THE CORRESPONDING CATEGORIES IN THE AID DATASET OF EACH SCENE

Index	Scene	Categories of AID Dataset	Count
0	Woodland	Forest, SparseResidential, Mountain	890
1	Water	Beach, Pond	820
2	Wasteland	Bareland, Desert	610
3	Cropland	Farmland	370
4	Building	Church, Commercial, Denseresidential, Industrial, School	1690
5	Grassland	Meadow, Baseballfield, Playground	872

III. EXPERIMENTS

A. Data

Due to the lack of public sharpness assessment datasets in the field of remote sensing image sharpness evaluation, the data used in the experiments were created based on the AID dataset (a public remote sensing scene classification dataset) [52]. The AID dataset has high spatial resolution and high image quality. It contains a total of 10 000 images, each sized 600 × 600. In order to highlight different scene data with different texture features, only all images for some categories in the AID dataset are collected. In order to highlight the effect of texture differences, inspired by the research [53] while combining China's remote sensing monitoring land use cover classification system, the dataset in this article contains six typical categories (woodland, water, wasteland, cropland, building, and grassland), which correspond to the six first-level types of the China's remote sensing monitoring land use cover classification system. The correspondence between each scene and the categories in the AID dataset is shown in Table I.

Blur and noise are important factors affecting the sharpness of remote sensing images. In order to produce data with different levels of sharpness, according to the standard of the research [54], Gaussian noise is added to the original remote sensing image while performing Gaussian blur processing. Specifically, Gaussian blur filters with window sizes of [11, 11] and variance σ_1 of 10, 20, 40, and 80 are used to obtain four remote sensing images with different degrees of blurring. At the same time, Gaussian noise with variance σ_2 of 5, 10, 20, and 40 are added to the four blurred images. Then, remote sensing images with four different degrees of blurs and noises can be obtained. Based on the original image and the four distorted images, the remote sensing images are

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TABLE II
SHARPNESS LEVEL OF REMOTE SENSING IMAGES

Degree of Distortion		Sharpness Level
$\sigma_1 = 0$	$\sigma_2 = 0$	Excellent
$\sigma_1 = 10$	$\sigma_2 = 5$	Good
$\sigma_1 = 20$	$\sigma_2 = 10$	Fair
$\sigma_1 = 40$	$\sigma_2 = 20$	Poor
$\sigma_1 = 80$	$\sigma_2 = 40$	Bad

classified into five sharpness levels, namely, excellent, good, fair, poor, and bad. The sharpness level of remote sensing images is shown in Table II.

A dataset comprising a total sample size of 26 260 remote sensing images is obtained after expanding the dataset samples and assigning sharpness-level labels. Each sample in this dataset is associated with both a scene category label and a sharpness level label. To effectively evaluate the accuracy of the model, the training set and the validation set are roughly divided according to the ratio of 1:1. The training set consists of 13 130 samples, while the validation set consists of 13 130 samples. In addition, rotation, cropping, and scaling operations are utilized to augment the sample size during training, thereby enhancing the network's robustness. Some representative images are shown in Table III.

B. Evaluation Metrics and Parameter Setting

In this article, the sharpness assessment method based on CS-Net outputs the quality level (excellent, good, fair, poor, and bad) rather than the absolute quality score. Therefore, the overall accuracy (OA) [55] is used to quantify sharpness assessment results. The metric of OA is defined as follows:

$$\mathrm{OA} = \frac{\sum_{k=1}^{N} x_{kk}}{\sum_{i=1}^{N}\sum_{j=1}^{N} x_{ij}} \tag{10}$$

where $\sum_{k=1}^{N} x_{kk}$ represents the sum of diagonal elements in the confusion matrix, that is, the total number of correctly classified samples, and $\sum_{i=1}^{N}\sum_{j=1}^{N} x_{ij}$ represents the sum of all elements in the confusion matrix, that is, the total number of samples.

The CS-Net is implemented using PyTorch. The model is trained by the adaptive moment estimation optimizer (momentum = 0.9 and weight decay = 0.0005). The epoch is set to 150, and the batch size is 64. The initial learning rate is set to 0.0005 and divided by ten every eight epochs.

C. Performance Comparison

The CS-Net is evaluated by comparing it with various mainstream classification methods on the dataset presented. The comparative methods encompass three traditional machine learning classification algorithms: local binary pattern (LBP) + support vector machine (SVM) [56], LBP + XGBoost [57], and LBP + light gradient boosting machine (LightGBM) [58]. These machine learning methods extract the LBP texture features from images and subsequently utilize SVM or decision tree models (XGBoost and LightGBM) for classification. In addition, six deep learning classification algorithms, namely, visual geometry group (VGG)-19 [59], ResNet-50 [60], WideResNet (WRN)-50-2 [61], ResNeXt-50 (32 × 4d) [62], squeeze-and-excitation (SE)-ResNet-50 [49], and Swin-transformer [44], are included. These deep learning methods perform classification by extracting deep-level features. Furthermore, six novel quality assessment networks, including RankIQA [40], MetaIQA [41], MANIQA [42], blind image quality assessment based on continual learning (BIQA_CL) [63], feature-level pseudo-reference information-based method (FPR) [64], and visual compensation restoration network (VCRNet) [65], have also been incorporated as comparative methods. The accuracy results of different methods on the dataset in this article are shown in Fig. 7, where the ordinate represents the accuracy and the abscissa represents the model's name.

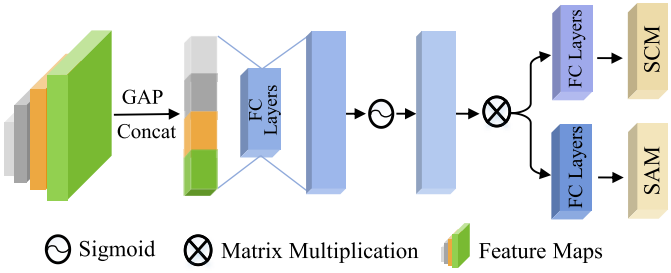


Fig. 6. Structure of the FFM.

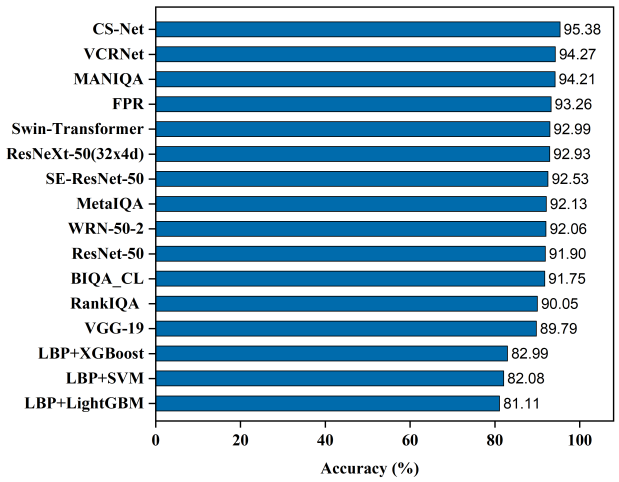


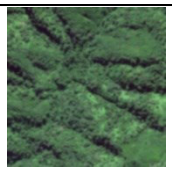
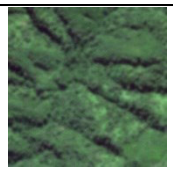
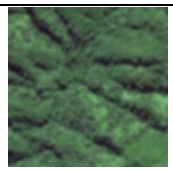
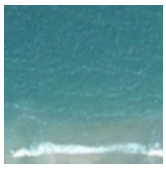
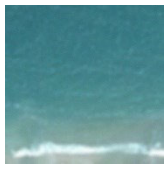
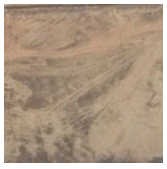
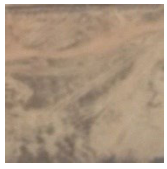
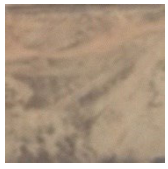
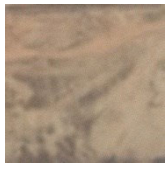
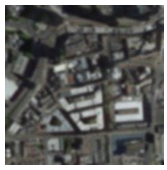
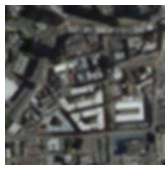
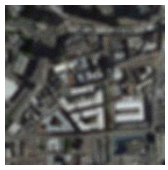
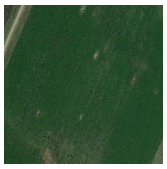
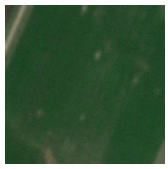
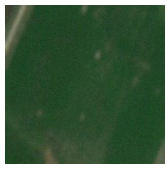
Fig. 7. Accuracy comparison of different methods on the dataset.

As shown in Fig. 7, the OA of the deep learning algorithms for sharpness assessment on the dataset are all higher than that of the traditional machine learning algorithms. In the experiment results from the mainstream machine learning methods, the OA of sharpness assessment for remote sensing images using LBP + LightGBM is 81.11%.

The OA of the LBP + XGBoost method and the LBP + SVM method are higher than that of the LBP + LightGBM method, reaching 82.99% and 82.08% respectively. In the experiment results from the deep learning methods, the OA of VGG-19 reaches 89.79%. The OAs of ResNet-50 and WRN-50-2 are relatively close, and they improve on VGG-19 by

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TABLE III
SOME REPRESENTATIVE IMAGES FROM THE DATASET

Scene Category \ Sharpness Level	Excellent	Good	Fair	Poor	Bad
Woodland					
Water					
Wasteland					
Cropland					
Building					
Grassland					

2.34% and 2.53%, respectively. Moreover, SE-ResNet-50 and ResNeXt-50 (32 × 4d) achieve well accuracy performance with 92.53% and 92.93%, respectively. Swin-transformer achieves an accuracy of 92.99% on the dataset, and it is also the backbone model used in the proposed method's feature extraction module. In the results of the six quality assessment models, VCRNet achieved the highest accuracy of 94.27%, while MANIQA, FPR, BIQA_CL, MetaIQA, and RankIQA achieved accuracies of 94.21%, 93.26%, 91.75%, 92.13%, and 90.05%, respectively. After comparing the results, it is evident that the CS-Net model proposed in this article exhibits the highest OA, at 95.38%, in sharpness assessment. This surpasses the second-highest, VCRNet, by about 1.2%. Furthermore, CS-Net demonstrates a significant advancement, approximately 14.9%, over the best-performing machine learning method, LBP + XGBoost, on the dataset. It also shows an improvement of nearly 2.6% compared to the well-performing deep learning method, Swin-transformer, on the same dataset.

To further demonstrate the effectiveness of the proposed method, sharpness assessment experiments were conducted based on the metrics (Tenengrad function, Laplacian function, Brenner function, SMD, and energy function) on the dataset. First, the scores of these metrics on the data were calculated. Then, based on the score ranking, the sharpness level corresponding to the image was determined. Finally, the accuracy was calculated by comparing it with the image's own sharpness level label. The statistical results of metric accuracy assessment in different scenes are shown in Fig. 8.

From Fig. 8, it can be observed that the overall accuracies of the sharpness assessment for the five metrics (Tenengrad function, Laplacian function, Brenner function, SMD, and energy function) are not high. The highest accuracy is achieved by the Brenner metric at only 56.53% in the "building" category. Furthermore, the accuracy of each metric varies significantly across different scenes. For example, the Tenengrad metric has an accuracy of 40.2% in the "building" category, while it is only 21.88% in the "wasteland" category. In summary, the

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TABLE IV
EXAMPLES OF DATA WITH DIFFERENT RESOLUTIONS FOR THE SAME AREA

Satellite	Resolution (m)	Woodland	Water	Cropland	Building
Sentinel-2	10				
	20				
LandSat-8	30				

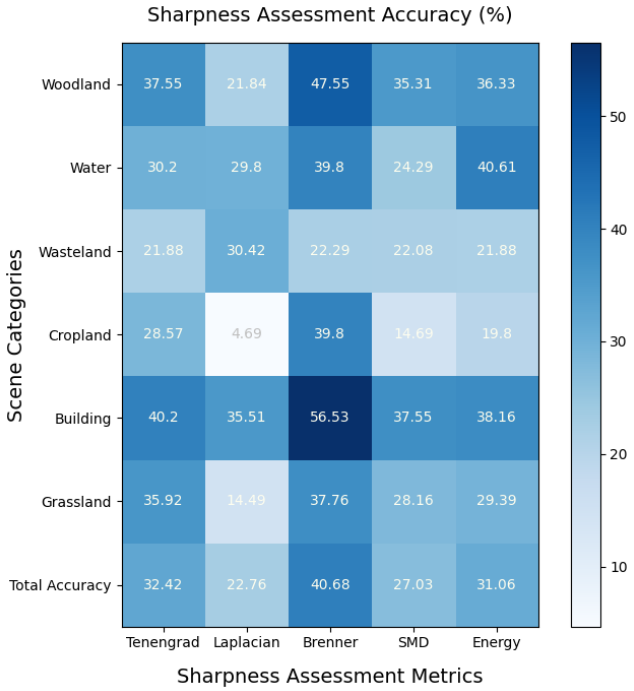


Fig. 8. Statistical results of accuracy assessment from metrics in different scenes.

assessment accuracy of these metrics is much lower than that of the method proposed in this article. Moreover, there is a significant accuracy gap across different categories, indicating insufficient reliability.

D. Experiments With Data of Different Resolutions

To further evaluate the performance of the proposed CS-Net on medium- and low-resolution data, the experiments used Sentinel-2 multispectral data (with resolutions of 10 and 20 m) and Landsat-8 multispectral data (with a 30-m resolution) covering the same area have been conducted.

Due to the scarcity of satellite remote sensing datasets for sharpness assessment tasks and the need to ensure consistency in texture features by using data of different resolutions covering the same area, finding suitable datasets becomes even more challenging. Therefore, this article describes the manual creation of a dataset (as shown in Table IV), which includes four types of scenes (woodland, water, cropland, and building). By expanding the dataset samples and assigning sharpness level labels, three datasets with different resolutions were obtained, each containing 400 remote sensing images. In the experiments based on these datasets, the training and validation sets were roughly divided in a 4:1 ratio. In addition, rotation, cropping, and scaling operations are utilized to augment the sample size during training.

The accuracy of sharpness assessment by various methods on data of different resolutions is presented in Table V. It is evident that as the resolution of the data diminishes, the assessment accuracy of most models correspondingly declines to varying degrees. This phenomenon can be attributed to two main reasons. First, a decrease in resolution reduces the texture information in the images, which may result in the models learning insufficient texture feature information. Second, the limited number of data samples used leads to the models' inability to learn sufficiently complex features, thus restricting their performance. However, it is still evident that the proposed method achieved the best assessment accuracy at resolutions of 10 and 20 m. Although it achieved the second-best accuracy at a 30-m resolution, it was very close to the highest accuracy. These results demonstrate the effectiveness of the proposed method in sharpness assessment on data of different resolutions.

E. Experiments on Typical Scene Data With Complex Objects

To test the effectiveness of the proposed CS-Net on typical scene data with complex objects, four types of typical scenes with complex objects were used. These four typical scenes are: the port scene (involving water, boats, buildings, impervious surfaces, and so on), the school scene (involving playgrounds,

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TABLE V
COMPARISON OF THE ACCURACY OF DIFFERENT MODELS ON DATA WITH DIFFERENT RESOLUTIONS

Method	Accuracy (%) at Different Resolutions (m)		
	10 m	20 m	30 m
VCRNet	94.02	92.79	**91.01**
MANIQA	94.23	92.87	90.45
FPR	93.48	92.01	90.22
Swin-Transformer	91.99	91.21	89.98
ResNeXt-50 (32x4d)	93.02	92.23	88.47
SE-ResNet-50	90.36	90.01	88.59
MetaIQA	93.15	91.07	89.64
WRN-50-2	91.17	90.12	88.32
ResNet-50	91.24	90.21	87.25
BIQA_CL	90.21	89.37	88.32
RankIQA	90.07	90.02	87.45
VGG-19	88.38	87.47	83.21
LBP + XGBoost	85.42	83.24	79.20
LBP + SVM	86.34	83.57	77.78
LBP + LightGBM	86.65	85.32	78.64
CS-Net	**94.73**	**92.98**	90.99

TABLE VI
EXAMPLES OF DATA WITH DIFFERENT RESOLUTIONS FOR THE SAME AREA

Scene Category	Port	School	Stadium	Urban elevated bridge
Image				

impervious surfaces, trees, buildings, and so on), the stadium scene (involving steel-frame venues, rubber tracks, grass, impervious surfaces, and so on), and the urban elevated bridge scene (involving buildings, impervious surfaces, vegetation, water, and so on). During the testing process, a total of 40 images were selected, ten from each scene. Each image was then subjected to degradation processing, resulting in a dataset of 200 images, each at five levels of sharpness. Some examples of the data are shown in Table VI.

The sharpness assessment results of the method proposed for four types of typical scene data are shown in Table VII. It can be observed that the lowest accuracy reached is 96.02%. This indicates that the proposed method is more effective in scenes containing complex objects. The higher effectiveness is likely because the proposed method is more sensitive to texture features, which significantly affect sharpness. Scenes with complex objects contain rich texture features, thereby improving the effectiveness of the sharpness assessment.

TABLE VII
PERFORMANCE COMPARISON OF SHARPNESS ASSESSMENT ON DIFFERENT SCENE DATA

Scene Category	Sharpness Assessment Accuracy (%)
Port	96.02
School	96.61
Stadium	96.34
Urban elevated bridge	97.63

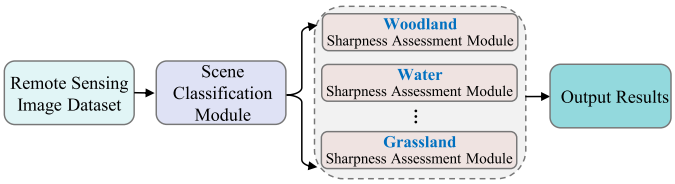


Fig. 9. Overall architecture of the comparative methodology.

TABLE VIII
PERFORMANCE COMPARISON OF SHARPNESS ASSESSMENT ON DIFFERENT SCENE DATA

Scene Category	Scene Classification Accuracy (%)	Sharpness Assessment Accuracy (%)
Woodland	98.49	95.87
Water	97.45	95.61
Wasteland	98.60	91.47
Cropland	98.25	89.27
Building	99.24	98.59
Grassland	98.22	88.76

IV. DISCUSSION

A. Impact Analysis of Scenes

To further analyze the impact of scene features, a comparative method was introduced. This method separates scene classification and sharpness assessment into two independent stages. First, the remote sensing image data are categorized into different scenes. Then, based on these categorized scenes, sharpness assessment is performed separately for each scene using six distinct sharpness evaluation modules, as illustrated in Fig. 9. Since six scene categories are considered in this article, six corresponding sharpness evaluation modules are designed, and they are not interrelated. The results obtained from scene classification serve as inputs to these modules, and the final evaluation results are obtained accordingly.

The accuracy of scene classification and the accuracy of sharpness assessment for each category are shown in Table VIII.

The table above illustrates two distinct characteristics. The first one is that the accuracy of scene classification is significantly higher than that of sharpness assessment, which reflects the greater difficulty of the sharpness assessment task

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TABLE IX
PERFORMANCE COMPARISON FROM EXPERIMENTS WITH DIFFERENT MODULES ON THE DATASET

Methods	Task schemes	Attention schemes	Fusion schemes	Loss schemes	Sharpness Level—Accuracy (%)					OA (%)
					Excellent	Good	Fair	Poor	Bad	
(a)	SAM	IAM	FFM	×	91.53	92.38	93.12	91.38	96.54	92.99
(b)	MTM	IAM	FFM	×	91.70	92.57	92.99	91.52	97.12	93.18
(c)	MTM	×	FFM	DBLF	92.78	93.96	94.35	93.43	97.79	94.46
(d)	MTM	IAM	×	DBLF	92.57	92.81	93.69	92.86	97.82	93.95
(e)	MTM	IAM	FFM	DBLF	**93.94**	**94.52**	**94.71**	**95.32**	**98.41**	**95.38**

in comparison to the scene classification task. The second characteristic is the significant disparity in sharpness assessment accuracy values among different scene categories, and this disparity can be attributed to two primary reasons. First, the scene classification accuracy is not 100%, thereby introducing some noise in the training of the sharpness assessment model. Second, there exists a significant difference in data volume among various categories (data volumes for each category are shown in Table I, the gap in data volume becomes even larger after rotating, cropping, and scaling the data). While the total data may be sufficient when considered as a whole, it is likely insufficient when input separately for certain categories during the sharpness assessment model training process.

B. Effectiveness of Modules

Furthermore, the validity of MTM, DBLF, IAM, and FFM is verified on the dataset of this article.

1) *Effectiveness of MTM:* To evaluate the effectiveness of MTM, an additional experiment was conducted for comparison, in which only sharpness assessment was performed without employing MTM. This experimental method is termed method (a) in Table IX. Method (e) in Table IX refers to the proposed CS-Net model, which comprises an SCM and an SAM. From the accuracy results in the table, it can be observed that the OA of method (e) has improved by 2.6% compared to method (a). This finding indicates the effectiveness of the approach proposed in this article, which takes into account scene texture features. The texture features of remote sensing images of different scenes have significant differences, which can be learned by the SCM. Then, these features learned can be shared with the SAM by the joint training, thus improving the prediction accuracy of the network.
2) *Effectiveness of IAM:* To evaluate the effectiveness of IAM, an additional comparative experiment was conducted, referred to as method (c) in Table IX. In this method, IAM was not used in the backbone network of CS-Net. The OA of method (c) without IAM was found to be 1% lower than that of the proposed method, reaching 94.46% according to the results in Table IX. IAM enables the neural network to focus more on important information and reduces the influence of useless background information. Specifically, the improved channel attention module can preserve more texture features of remote sensing images, while its enhanced spatial attention module helps highlight the informative features of meaningful target locations. Therefore, IAM contributes to the overall performance improvement of the CS-Net network model.
3) *Effectiveness of FFM:* To evaluate the effectiveness of FFM, a comparative experiment was conducted, denoted as method (d) in Table IX. In this method, the features extracted by the CS-Net backbone network were not subjected to fusion and were directly input to the MTM for training. From the OA results in Table IX, it can be observed that method (d) achieved an OA of 93.95%, which is 1.5% lower than the OA of the proposed method. FFM helps to compensate for the resolution and positional information lost during multiple pooling of FMs, enabling the network to better correspond to semantic and positional information. The OA comparison results demonstrate that FFM has played a role in improving the accuracy of the remote sensing image sharpness assessment task.
4) *Effectiveness of DBLF:* To evaluate the effectiveness of DBLF, an additional comparative experiment, referred to as method (b) in Table IX, was conducted. In this comparative experiment, the weights of the SCM and the SAM were fixed at empirical values of 0.2 and 0.8, respectively, followed by end-to-end training. From the accuracy results in Table IX, it can be observed that the proposed method (e) achieved an OA improvement of 2.4% compared to method (b). This improvement is attributed to the proposed method introduced by DBLF based on the gradient balancing strategy. DBLF dynamically adjusts the weights of the two task losses in the total loss during the training process based on the prediction performance of the SCM and the SAM. By decreasing the weight of the task with better accuracy and increasing the weight of the task with poorer accuracy, the network becomes more focused on the more

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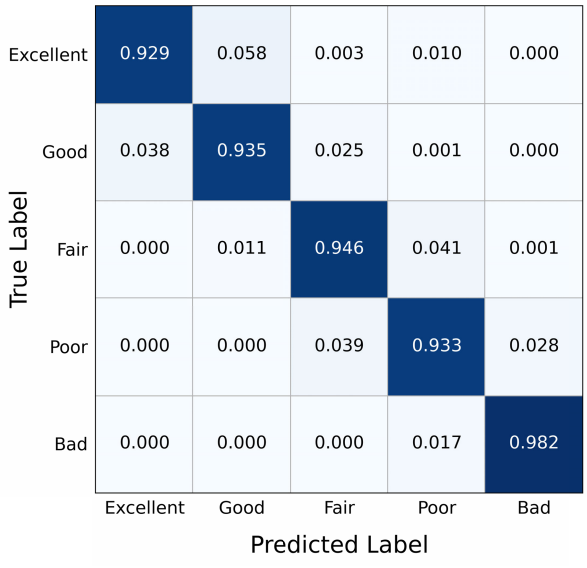


Fig. 10. Confusion matrix of the sharpness evaluation results for CS-Net.

challenging task. As evident from Table IX, in the multitask training of this article, the sharpness assessment task is more difficult. Therefore, DBLF helps the overall model to focus more on the sharpness assessment task.

C. Correlation Analysis

To further investigate the correlation among the five defined sharpness levels in this article and analyze the accuracy of CS-Net in classifying each level, a confusion matrix was computed and visually presented in Fig. 10. The horizontal axis represents the predicted sharpness levels, while the vertical axis represents the actual sharpness levels.

From Fig. 10, it can be observed that CS-Net achieved the highest accuracy of 0.982 in the bad level. Even at the excellent level, the accuracy remains the lowest, but it still reaches 0.929. CS-Net exhibited low misclassification rates across all five sharpness levels, with the highest misclassification occurring in the excellent and good levels, with a misclassification rate of 0.058. In conclusion, CS-Net demonstrated a favorable overall performance in the sharpness assessment of the provided remote sensing image dataset in this study.

D. Computational Cost Discussion

To further discuss the efficiency of the model, a comparison of computational cost was conducted between the proposed model and other deep learning models (MANIQA, Swin-transformer, ResNeXt-50 (32×4d), SE-ResNet-50, MetaIQA, WRN-50-2, ResNet-50, RankIQA, VGG-19, FPR, BIQA_CL, and VCRNet), with the results shown in Table X. Combined with the accuracy results shown in Fig. 7, although the number of parameters and the complexity of the proposed CS-Net are higher than most models, it is still lower than several advanced models (MANIQA and VCRNet). More importantly, CS-Net achieves state-of-the-art predictive performance. Consequently, it can be deduced that the proposed CS-Net achieves state-of-the-art predictive performance while simultaneously maintaining comparable computational complexity.

TABLE X
COMPUTATIONAL COST OF DIFFERENT MODELS. "PARAMS" REFERS TO THE NUMBER OF NETWORK PARAMETERS, WHILE "FLOPS" DENOTES THE NUMBER OF FLOATING-POINT OPERATIONS, CALCULATED ON IMAGES WITH A SPATIAL RESOLUTION OF 224×224 PIXELS

Method	Params (/M)	FLOPs (/G)
CS-Net	217.08	33.39
MANIQA	243.82	55.96
Swin-Transformer	195.29	30.95
ResNeXt-50 (32x4d)	25.00	4.24
SE-ResNet-50	29.14	3.91
MetaIQA	13.24	1.83
WRN-50-2	68.88	11.45
ResNet-50	25.56	4.13
RankIQA	134.28	15.52
VGG-19	143.67	19.69
FPR	16.52	78.35
BIQA_CL	44.63	22.35
VCRNet	473.59	88.37

V. CONCLUSION

In this article, a sharpness assessment method for remote sensing images based on a deep multibranch network considering scene features is proposed. The integration of scene classification and sharpness evaluation is achieved through a specially designed MTM. The accuracy of sharpness assessment is enhanced by leveraging shared features that capture the correlation between scenes and sharpness. To effectively utilize feature information at different scales, the IAM is used and an FFM is introduced. Moreover, a total loss function employing the DBLF based on the gradient balancing strategy is formulated to mitigate the imbalance phenomenon observed in multimodule synchronous training, where one task shows excellent prediction performance, while other tasks suffer. Experimental results, including a comparative analysis with other methods, showcase the potential of the proposed method in the field of remote sensing image sharpness assessment.

Nevertheless, there is still room for further refinement in the sharpness assessment in remote sensing imagery. For instance, there are relatively few publicly available datasets that can be used for the sharpness assessment of remote sensing images, which limits the development of comprehensive sharpness assessment experiments. Therefore, a dataset with a larger sample size for sharpness assessment of remote sensing images will be established in the future, which will be fostering the enhancement of sharpness assessment algorithms. Furthermore, more refined scene categories covering practical applications will be analyzed, further exploring the correlation between the image sharpness of remote sensing images and scene features.

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